

FL Simulation Results

File: test_B5

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 10
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 20.0% (0.20)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 20.0% (0.20) of data for 15 rounds with 10 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8728
Clean + DP Protection Accuracy: 0.8727
Poisoned Accuracy: 0.8380
Poisoned + DP Protection Accuracy: 0.8379
Drop (Clean - Poisoned): 0.0349
Drop (Clean - Init): 0.0800
Drop (Clean DP - Init): 0.0799
Drop (Poisoned - Init): 0.0452
Drop (Poisoned DP - Init): 0.0451
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8791
Clean Recall: 0.8728
Clean F1 Score: 0.8723

Clean DP Precision: 0.8791
Clean DP Recall: 0.8727
Clean DP F1 Score: 0.8722

Poisoned Precision: 0.8465
Poisoned Recall: 0.8380
Poisoned F1 Score: 0.8370

DP Protection Precision: 0.8476
DP Protection Recall: 0.8379
DP Protection F1 Score: 0.8372

Summary

FL Simulation Complete
Task: e91ffb5f-7289-4d26-99d7-d6a82fc6d88c
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8728
Clean DP Accuracy: 0.8727
Poisoned Accuracy: 0.8380
Data Poison Protection Accuracy: 0.8379
Drop (Clean - Poisoned): 0.0349
Drop (Clean - Init): 0.0800
Drop (Clean DP - Init): 0.0799
Drop (Poisoned - Init): 0.0452
Drop (Poisoned_DP - Init): 0.0451
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8791
Clean Recall: 0.8728
Clean F1 Score: 0.8723
Clean DP Precision: 0.8791
Clean DP Recall: 0.8727
Clean DP F1 Score: 0.8722
Poisoned Precision: 0.8465
Poisoned Recall: 0.8380
Poisoned F1 Score: 0.8370
DP Protection Precision: 0.8476
DP Protection Recall: 0.8379
DP Protection F1 Score: 0.8372